

Supplemental Material for TSUSC-2026-07-0200

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I. ADDITIONAL EXPERIMENTAL DETAILS

This supplement provides additional calibration, per-seed statistical, online deployment, energy, and test-set results for “Shielded Q-learning for Thermally-Safe, Energy-Aware Multi-Model Intrusion Detection Scheduling on Lightweight Edge Devices.” Calibration, Q training, online hardware experiments, physical-energy measurement, and statistical analysis were completed before the official UNSW-NB15 test was accessed. The test set was then evaluated once using fixed preprocessing and detector thresholds.

II. DEVICE-SPECIFIC GUARD CALIBRATION

Both devices use independent fit, margin, and held-out coverage splits over 72 action/workload/background cells. Each split contains 2,880 windows. No coverage observation is used to refit a predictor or change a margin.

TABLE I
ACCEPTED HELD-OUT CALIBRATION COVERAGE.

Device	n	Latency	Thermal	Joint
Pi 3B+	2880	2879/2880	2880/2880	2879/2880
Pi 4B 8GB	2880	2880/2880	2877/2880	2877/2880

For Pi 4B, by-action joint coverage is 1.0000 for FISVDD and TinyDL, 0.9972 for LUCID, and 0.99861 for OI-SVDD+AS-ELM. Three thermal-side uncovered observations are retained. The lowest 40-window cell coverage is 0.975. The Pi 3B+ calibration likewise retains its single uncovered joint observation and is reported without rounding to 1.0.

III. PI 3B+ BINDING-RESOURCE COMPARISON

The accepted binding-resource campaign contains 6 policies $\times$ 10 matched seeds $\times$ 600 windows = 36,000 windows. The guard has exactly three admissible actions in every window and excludes LUCID. No fallback occurs.

For the primary FSSQL-R minus Safe-Greedy comparison, exact paired sign-flip tests give $p = 0.00195$ for F1, cumulative reward, and switching cost, and $p = 0.00586$ for mean E2E latency. The mean controller-overhead difference is -0.0113 ms ($p = 0.23242$). Both policies have zero realized violations.

The single Threshold violation occurs at seed 4109, window 111. TinyDL is selected with a guarded predicted latency of 5.147 ms and 24.297 ms headroom, but complete E2E latency is 52.522 ms. Detector inference is 0.446 ms and total controller time is 50.414 ms; no switching, thermal, frequency, or undervoltage anomaly is present. This event motivates the distinction between action-level guard calibration and complete pipeline latency.

IV. Q-LEARNING STABILITY DIAGNOSTICS

The Q-learning implementation uses 1280 states and 5120 state-action pairs, $\gamma = 0.95$, $\alpha = \min\{0.5, (1 + N(s, a))^{-0.6}\}$, epsilon linearly decayed from 1.0 to 0.05 by window 4500, a replay capacity of 512, and one online plus four replay updates per training window.

FSSQL-R visits 29.7 states/96.3 state-actions by window 600 and 34.9/118.9 by window 6000. Its greedy-policy change rate falls from 0.006875 to 0.000156. These diagnostics support late-budget behavioral stability on visited states, not mathematical convergence over the entire state space.

V. PI 4B ONLINE AND PHYSICAL-ENERGY RESULTS

The Pi 4B online evaluation contains 60 policy-condition-seed cells and 36,000 windows. Two runs had incomplete contemporaneous POWER-Z traces and were repeated once under the same fixed configuration. Analyses use only complete runs, and measurements are not mixed across attempts.

For seed 5102, FSSQL-R selects FISVDD in 582/600, 592/600, and 588/600 windows under B0/B1/B2, respectively (97.0–98.7%). The other four seeds have mean FPR 0.1384, 0.1380, and 0.1517 across B0/B1/B2. Thus the aggregate mean and SD summarize a bimodal seed pattern poorly; the main paper therefore reports FPR as median [min,max]. With only five seeds, the observed one-of-five collapse is not interpreted as a population failure probability.

Safe-Greedy selects FISVDD in all 15 Pi 4B formal runs and therefore yields FPR=1.0000 and TN=0 throughout. This explains why the FSSQL-R-minus-Safe-Greedy FPR differences are large and negative. Against the stronger Threshold controller, the FSSQL-R-minus-Threshold mean FPR differences are +0.2211, +0.2479, and +0.2619 for B0/B1/B2, with bootstrap 95% CIs [+0.0249,+0.5384], [+0.0323,+0.5928], and [+0.0581,+0.5896]. The corresponding F1 mean differences are -0.0256, -0.0261, and -0.0406, and their bootstrap intervals cross zero. With only five pairs, these FPR differences do not reach 0.05 under the paired t or Wilcoxon tests; they are reported as directional evidence rather than a universal significance result.

The Pi 4B guard is nonbinding in the accepted online conditions: the mean safe-set size is four for every policy/condition. Mean controller time across all conditions is 3.919 ms for FSSQL-R, 3.885 ms for Safe-Greedy, 4.277 ms for Static-Light, and 3.946 ms for Threshold. Mean selected-detector inference is 0.415, 0.426, 0.648, and 0.299 ms, respectively. This confirms that controller work is a material part of online cost.

VI. ONE-SHOT OFFICIAL TEST

After model, preprocessing, threshold, and scheduling choices were fixed, all 82,332 rows of the official UNSW-

TABLE II
Pi 3B+ MATCHED-SEED AGGREGATE RESULTS.

Policy	F1	Cum. reward	Mean E2E ms	P99 E2E ms	Controller ms	Switches	Switch cost	Viol.
FSSQL-R	0.8523	326.06	8.3149	9.5477	5.9033	267.4	5.348	0
Safe-Greedy	0.8708	350.34	7.9775	8.9389	5.9146	0.0	0	0
Threshold	0.8704	350.04	7.9704	9.1303	5.9035	0.6	0.012	1
Unshielded-Q	0.8523	329.35	10.1141	16.3117	5.8988	221.9	4.438	0
Round-Robin	0.8148	288.04	10.6571	18.2874	5.8860	599.0	11.980	0
Static-Light	0.7203	203.31	8.2550	9.2435	5.8915	0.0	0	0

TABLE III
LEARNING CHECKPOINTS (MEAN ACROSS 10 TRAINING SEEDS).

Policy	R@600	U@600	R@6000	U@6000
FSSQL-R	0.483742	0.575863	0.526027	0.610795
Unshielded-Q	0.488472	0.581728	0.518533	0.603689

NB15 test were evaluated once. No retraining or threshold selection followed test access.

VII. STATEFUL EXTENSION STOP RESULT

A planned Pi 3B+ stateful-transition extension measured action-dependent service timing under B0/B1/B2, but the pre-defined service-model check could not simultaneously demonstrate backlog accumulation in B2/B1 and recovery in B0 without introducing a new arrival rate or changing the service criterion. The extension was therefore stopped before policy runs. This result does not provide evidence for or against Q-learning superiority.

TABLE IV
FSSQL-R MINUS SAFE-GREEDY PAIRED DIFFERENCES ON P1 4B (FIVE MATCHED SEEDS PER CONDITION).

Load	Metric	Mean diff.	95% bootstrap CI	Paired t p	Exact Wilcoxon p	d_z
B0	F1	0.1209	[0.0664, 0.1535]	0.0114	0.0625	1.984
B0	FPR	-0.7064	[-0.9061, -0.3924]	0.0111	0.0625	-1.995
B0	Cum. reward	129.4	[72.0, 162.6]	0.0106	0.0625	2.026
B0	P99 E2E ms	0.323	[0.091, 0.548]	0.0562	0.1250	1.191
B0	Total energy J	-0.471	[-13.829, 12.078]	0.9516	1.0000	-0.029
B1	F1	0.1172	[0.0596, 0.1515]	0.0152	0.0625	1.821
B1	FPR	-0.6940	[-0.8986, -0.3503]	0.0151	0.0625	-1.824
B1	Cum. reward	124.8	[64.0, 159.4]	0.0147	0.0625	1.838
B1	P99 E2E ms	-0.129	[-0.560, 0.300]	0.6341	0.6250	-0.230
B1	Total energy J	-3.427	[-7.088, -0.695]	0.1360	0.0625	-0.833
B2	F1	0.1141	[0.0602, 0.1489]	0.0136	0.0625	1.882
B2	FPR	-0.6891	[-0.8999, -0.3574]	0.0131	0.0625	-1.904
B2	Cum. reward	122.0	[65.2, 157.8]	0.0126	0.0625	1.923
B2	P99 E2E ms	0.033	[-0.506, 0.700]	0.9281	0.8125	0.043
B2	Total energy J	-1.210	[-11.564, 10.608]	0.8533	0.6250	-0.088

TABLE V
FSSQL-R SEED-LEVEL P1 4B BEHAVIOR. SEED 5102 IS A REPEATED HIGH-FPR MODE ACROSS ALL THREE BACKGROUND CONDITIONS.

Seed	B0 FPR	B1 FPR	B2 FPR	B0 F1	B1 F1	B2 F1
5101	0.2233	0.1869	0.1818	0.9436	0.9560	0.9511
5102	0.9144	0.9780	0.9476	0.8285	0.8245	0.7967
5103	0.0674	0.0444	0.0378	0.9421	0.9624	0.9631
5104	0.1917	0.1931	0.1882	0.9565	0.9533	0.9641
5105	0.0714	0.1276	0.1990	0.9467	0.9394	0.9357

TABLE VI
OFFICIAL TEST CONFUSION MATRICES AND RANKING METRICS.

Detector	TP	FP	FN	TN	Precision	Recall	F1	FPR	AUROC	PR-AUC
FISVDD	45332	37000	0	0	0.5506	1.0000	0.7102	1.0000	0.5351	0.6219
LUCID	45211	19188	121	17812	0.7020	0.9973	0.8240	0.5186	0.8621	0.8469
TinyDL	44414	12514	918	24486	0.7802	0.9797	0.8686	0.3382	0.9704	0.9784
OI-SVDD+AS-ELM	45218	15581	114	21419	0.7437	0.9975	0.8521	0.4211	0.9513	0.9584